

Lectures 5–7: Wick’s Theorem and the Semicircle Law

Pair partitions, counting paths, and the Stieltjes transform

OUTLINE L5 Wick’s theorem $\rightarrow$ the Gaussian Wigner matrix $\rightarrow \|Mu\|^2$ for a fixed direction
 L6 The semicircle law $\rightarrow$ the strategy $\rightarrow \mathbb{E} \operatorname{tr} M^2, \mathbb{E} \operatorname{tr} M^4$ by hand $\rightarrow$ traces are sums over paths $\rightarrow$ the same two moments by paths $\rightarrow$ every moment
 L7 The empirical spectral measure $\rightarrow$ the Stieltjes transform $\rightarrow$ Schur complement $\rightarrow m^2 + zm + 1 = 0 \rightarrow$ the density

HOW TO READ Sections marked * are not lectured. They are complete, they are examinable, and they carry most of the exercises. Exercises are typed A, B and C; each week’s quiz has one of each.

1 Wick’s theorem, and Gaussian random matrices

Q. What can be computed about a random matrix without touching its eigenvalues?

Every moment of a Gaussian vector is a counting problem, and the entries of a symmetric Gaussian matrix are a Gaussian vector. So anything polynomial in the entries — the length of Mu for a fixed vector u , for instance — can be computed exactly.

1.1 Pair partitions

Definition 1.1 (Pair partition). A *pair partition* of $\{1, \dots, 2q\}$ is a partition into q blocks of size two.

Theorem 1.2 (Wick). Let $X \sim \mathcal{N}(0, \Sigma)$ on $\mathbb{R}^d$ and $a_1, \dots, a_m \in \{1, \dots, d\}$. If m is odd then $\mathbb{E}[X_{a_1} \cdots X_{a_m}] = 0$. If $m = 2q$ then

$$\mathbb{E}[X_{a_1} X_{a_2} \cdots X_{a_{2q}}] = \sum_{\text{pair partitions } P \text{ of } \{1, \dots, 2q\}} \prod_{\{r, s\} \in P} \Sigma_{a_r a_s}$$

Proof. By Corollary 1.7 of Lectures 1–2 every linear functional of X is a one-dimensional Gaussian, so X has moments of every order, φ_X is infinitely differentiable, and its derivatives may be taken inside the expectation. Since $\partial_{\xi_j} e^{-i\langle x, \xi \rangle} = -ix_j e^{-i\langle x, \xi \rangle}$, multiplication by X_j acts on the transform as $i \partial_{\xi_j}$:

$$\mathbb{E}[X_{a_1} \cdots X_{a_m}] = i^m \partial_{\xi_{a_1}} \cdots \partial_{\xi_{a_m}} \varphi_X(\xi) \Big|_{\xi=0}. \quad (1)$$

By Definition 1.3 of Lectures 1–2, $\varphi_X(\xi) = \exp(-\frac{1}{2} \xi^\top \Sigma \xi)$ — for every $\Sigma \succeq 0$, degenerate cases included — and differentiating it once,

$$\partial_{\xi_i} \varphi_X(\xi) = - \sum_j \Sigma_{ij} \xi_j \varphi_X(\xi). \quad (2)$$

Apply $\partial_{\xi_{a_2}} \cdots \partial_{\xi_{a_m}}$ to (2) taken at $i = a_1$. The factor ξ_j is linear, so Leibniz leaves terms of two kinds: those in which every derivative falls on φ_X , which still carry a factor ξ_j ; and, for each

$s \in \{2, \dots, m\}$, the one in which $\partial_{\xi_{a_s}}$ falls on ξ_j and produces δ_{ja_s} . Setting $\xi = 0$ kills the first kind, leaving

$$\partial_{\xi_{a_1}} \cdots \partial_{\xi_{a_m}} \varphi_X|_0 = - \sum_{s=2}^m \Sigma_{a_1 a_s} \partial_{\{\xi_{a_r}: r \neq 1, s\}} \varphi_X|_0.$$

Rewrite both sides with (1) and multiply through by i^m ; since $i^2 = -1$ the two minus signs cancel and

$$\mathbb{E}[X_{a_1} \cdots X_{a_m}] = \sum_{s=2}^m \Sigma_{a_1 a_s} \mathbb{E}\left[\prod_{r \neq 1, s} X_{a_r}\right] : \quad (3)$$

pair the first factor with each of the others in turn, and recurse. Induct on m . Both sides vanish for $m = 1$, and for $m = 2$ the claim is the definition of Σ . For $m > 2$, (3) drops the number of factors by two, so parity is preserved: an odd product reduces to $\mathbb{E}[X_a] = 0$, and an even one to a sum over pair partitions of the remaining $2q - 2$ indices, each prefixed by $\Sigma_{a_1 a_s}$ recording the block containing 1. Summing over s enumerates every pair partition of $\{1, \dots, 2q\}$ once. ■

Remark 1.3. Reading (3) with $F(X) = X_{a_2} \cdots X_{a_m}$ and noting that $\partial_j F$ deletes the factor X_j gives

$$\mathbb{E}[X_i F(X)] = \sum_j \Sigma_{ij} \mathbb{E}[\partial_j F(X)],$$

Gaussian integration by parts, which for $d = 1$ and $\Sigma = 1$ is $\mathbb{E}[gF(g)] = \mathbb{E}[F'(g)]$ — Lemma 1.10 of Lectures 1–2, and Stein’s lemma in Lecture 4. It is Exercise 2.3 of Lectures 3–4; here it falls out of Wick’s recursion for free.

1.2 Three examples

Corollary 1.4. *There are $(2q - 1)!! = (2q - 1)(2q - 3) \cdots 3 \cdot 1$ pair partitions of $2q$ points, and $\mathbb{E}[g^{2q}] = (2q - 1)!!$ for $g \sim \mathcal{N}(0, 1)$.*

Proof. Take $d = 1$ and $\Sigma = 1$ in Theorem 1.2: every product of covariances equals 1, so the moment is the number of pair partitions. That number is $(2q - 1)!!$ because the point 1 has $2q - 1$ choices of partner and deleting that block leaves $2q - 2$ points. ■

Example 1.5 (Four indices). For $q = 2$ the three pair partitions of $\{1, 2, 3, 4\}$ are $\{12\}\{34\}$, $\{13\}\{24\}$, $\{14\}\{23\}$, so

$$\mathbb{E}[X_1 X_2 X_3 X_4] = \Sigma_{12} \Sigma_{34} + \Sigma_{13} \Sigma_{24} + \Sigma_{14} \Sigma_{23}.$$

Setting all four indices equal recovers $\mathbb{E}[X_1^4] = 3\Sigma_{11}^2$.

Example 1.6 (A quadratic form). Let $X \sim \mathcal{N}(0, \Sigma)$ on $\mathbb{R}^d$ and let A be symmetric. Then $\mathbb{E}[X^\top A X] = \sum_{ij} A_{ij} \mathbb{E}[X_i X_j] = \text{tr}(A\Sigma)$, no Wick needed. For the second moment,

$$\mathbb{E}[(X^\top A X)^2] = \sum_{ijkl} A_{ij} A_{kl} \mathbb{E}[X_i X_j X_k X_l] = \sum_{ijkl} A_{ij} A_{kl} (\Sigma_{ij} \Sigma_{kl} + \Sigma_{ik} \Sigma_{jl} + \Sigma_{il} \Sigma_{jk})$$

by Example 1.5. The first term is $(\text{tr } A\Sigma)^2$; the other two are each $\text{tr}(A\Sigma A\Sigma)$, by symmetry of A and Σ . Hence

$$\text{Var}(X^\top A X) = 2 \text{tr}((A\Sigma)^2), \quad \text{and in particular} \quad \text{Var}(\|X\|^2) = 2 \text{tr } \Sigma^2.$$

1.3 The Gaussian Wigner matrix

Definition 1.7 (Gaussian Wigner matrix). Let A be $N \times N$ with i.i.d. $\mathcal{N}(0, 1)$ entries and put

$$M = \frac{1}{\sqrt{2N}}(A + A^\top).$$

Then M is symmetric, its entries $\{M_{ij}\}_{i \leq j}$ are independent centered Gaussians with $\mathbb{E}M_{ij}^2 = \frac{1}{N}$ for $i < j$ and $\mathbb{E}M_{ii}^2 = \frac{2}{N}$, and for *all* four indices

$$\mathbb{E}[M_{ij}M_{kl}] = \frac{1}{N}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}). \quad (4)$$

This ensemble is the *Gaussian orthogonal ensemble*, or GOE, for the reason in Remark 1.8. Everything below is (4) and Theorem 1.2; the construction itself never appears again. The two delta terms are the two ways of matching $\{i, j\}$ with $\{k, l\}$, and they are there because $M_{ij} = M_{ji}$: an entry may be met twice under either name. The diagonal variance $\frac{2}{N}$ comes out of the symmetrization (Exercise 1.9), and it is exactly what makes (4) hold at $i = j = k = l$ without a special case. The N diagonal entries are a vanishing fraction of the N^2 .

Remark 1.8 (Why “orthogonal”). For any fixed orthogonal O , $OAO^\top$ again has i.i.d. $\mathcal{N}(0, 1)$ entries — each entry is a unit linear functional of A , and Theorem 1.5 of Lectures 1–2 makes the collection jointly Gaussian with the same covariance — so $OMO^\top \stackrel{d}{=} M$. The law of the spectrum therefore cannot depend on the basis. This is the matrix form of the rotational invariance of Corollary 1.6 of Lectures 1–2.

Exercise 1.9 (TYPE A). Verify Definition 1.7 from the construction: with A having i.i.d. $\mathcal{N}(0, 1)$ entries and $M = \frac{1}{\sqrt{2N}}(A + A^\top)$, compute $\mathbb{E}[M_{ij}M_{kl}]$ directly and obtain (4). Check the case $i = j = k = l$ separately and confirm that the diagonal convention $\mathbb{E}M_{ii}^2 = \frac{2}{N}$ is forced, not chosen.

1.4 One direction at a time

Before the eigenvalues, a question with the same shape but no linear algebra in it: fix a unit vector u and ask how long Mu is.

Proposition 1.10. *Let M be as in Definition 1.7 and let $u \in \mathbb{R}^N$ with $\|u\| = 1$. Then Mu is a centered Gaussian vector with covariance*

$$C = \mathbb{E}[(Mu)(Mu)^\top] = \frac{1}{N}(I + uu^\top),$$

and consequently

$$\mathbb{E}\|Mu\|^2 = 1 + \frac{1}{N}, \quad \text{Var}\|Mu\|^2 = \frac{2}{N} + \frac{6}{N^2}.$$

Proof. Each coordinate $(Mu)_i = \sum_j M_{ij}u_j$ is a linear functional of the entries of M , so by Theorem 1.5 of Lectures 1–2 the vector Mu is centered Gaussian. Its covariance is (4) contracted with u twice:

$$\mathbb{E}[(Mu)_i(Mu)_k] = \sum_{j,l} u_j u_l \cdot \frac{1}{N}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) = \frac{1}{N}(\delta_{ik}\|u\|^2 + u_i u_k),$$

which is C . Then $\mathbb{E}\|Mu\|^2 = \text{tr } C = \frac{1}{N}(N + 1)$. For the variance, Example 1.6 with $A = I$ and $\Sigma = C$ gives $\text{Var}\|Mu\|^2 = 2 \text{tr } C^2$, and since $u^\top u = 1$,

$$(I + uu^\top)^2 = I + 2uu^\top + u(u^\top u)u^\top = I + 3uu^\top, \quad \text{tr } C^2 = \frac{1}{N^2}(N + 3). \quad \blacksquare$$

Remark 1.11. Two things are worth reading off. First, the answer does not depend on u — it cannot, because $OMO^\top \stackrel{d}{=} M$ by Remark 1.8, so the law of $\|Mu\|^2$ is the same in every direction. Second, the standard deviation is $\sqrt{2/N}$ against a mean of 1: for large N a Gaussian Wigner matrix maps *every* unit vector to a vector of length very close to 1. That is already a statement about eigenvalues, since $\|Mu\|^2 = u^\top M^2 u$ is a weighted average of the λ_i^2 with weights $\langle u, \phi_i \rangle^2$ summing to 1, where ϕ_i are the eigenvectors. Averaging over directions instead of fixing one leads to $\frac{1}{N} \text{tr } M^2$, which is taken up in §2.3.

Exercise 1.12 (TYPE A). Let $u, v \in \mathbb{R}^N$ be unit vectors. Show from (4) alone that $u^\top Mv$ is centered Gaussian with

$$\text{Var}(u^\top Mv) = \frac{1}{N}(1 + \langle u, v \rangle^2), \quad \text{and} \quad \text{Cov}(u^\top Mu, v^\top Mv) = \frac{2}{N} \langle u, v \rangle^2.$$

In particular $u^\top Mu$ has variance $\frac{2}{N}$, and for $u \perp v$ the variable $u^\top Mv$ has variance $\frac{1}{N}$: the diagonal and off-diagonal variances of Definition 1.7, seen in any orthonormal basis containing u and v . Remark 1.8 predicted exactly this.

Exercise 1.13 (TYPE B). Let $u, v \in \mathbb{R}^N$ be unit vectors. Show that

$$\text{Cov}(\|Mu\|^2, \|Mv\|^2) = \frac{2}{N^2} (1 + (N+2)\langle u, v \rangle^2),$$

and check that $u = v$ recovers Proposition 1.10. Mu and Mv are jointly Gaussian; compute their cross-covariance $\mathbb{E}[(Mu)(Mv)^\top]$ from (4), then apply Theorem 1.2 to $\mathbb{E}[(Mu)_i^2 (Mv)_k^2]$. For $u \perp v$ the correlation of the two lengths is $\frac{1}{N+3}$: a Gaussian Wigner matrix stretches orthogonal directions almost independently.

1.5 * Cumulants

Not lectured. Theorem 1.2 is the Gaussian instance of a formula that holds for any random vector with moments. This section states that formula; the exercises develop the properties that make it useful.

Definition 1.14 (Joint cumulants). For $Y_1, \dots, Y_m$ with all moments finite,

$$\kappa(Y_1, \dots, Y_m) = \frac{\partial^m}{\partial t_1 \dots \partial t_m} \log \mathbb{E}[e^{t_1 Y_1 + \dots + t_m Y_m}] \Big|_{t=0}.$$

Moments are derivatives of the transform; cumulants are derivatives of its logarithm. The m th cumulant of a single variable is $\kappa(Y, \dots, Y)$ with m arguments; when the argument is itself a sum, keeping the m slots visible is what makes multilinearity usable.

Stated without proof: the moment–cumulant formula. For any $Y_1, \dots, Y_m$ with moments,

$$\mathbb{E}[Y_1 Y_2 \dots Y_m] = \sum_{\pi \in \mathcal{P}(m)} \prod_{B \in \pi} \kappa(Y_i : i \in B),$$

where $\mathcal{P}(m)$ is the set of *all* partitions of $\{1, \dots, m\}$ into non-empty blocks, of which there are 1, 2, 5, 15, 52, ...

For $m = 2$ this reads $\mathbb{E}[Y_1 Y_2] = \mathbb{E}Y_1 \mathbb{E}Y_2 + \text{Cov}(Y_1, Y_2)$, the two partitions of $\{1, 2\}$. If the Y_i are coordinates of a centered Gaussian vector, every joint cumulant of order other than two vanishes

(Exercise 1.16), so a partition contributes nothing unless all of its blocks have size two. Those are the pair partitions, and the moment–cumulant formula becomes Theorem 1.2. Of the fifteen partitions of a four-element set, three survive, which is Example 1.5.

Remark 1.15. For a non-Gaussian vector the larger blocks do not vanish, $\log \mathbb{E}[e^{\langle t, X \rangle}]$ is no longer a polynomial, and the recursion (3) acquires a tail that never terminates. Cumulants then measure the distance from Gaussian, and because mixed cumulants of independent variables vanish, they are the natural bookkeeping for sums of independent variables.

Exercise 1.16 (TYPE B). Prove the following directly from Definition 1.14.

- (a) **Low orders.** $\kappa(Y) = \mathbb{E}Y$ and $\kappa(Y_1, Y_2) = \text{Cov}(Y_1, Y_2)$.
- (b) **Multilinearity.** κ is symmetric in its arguments and linear in each: $\kappa(aY + bZ, Y_2, \dots, Y_m) = a\kappa(Y, Y_2, \dots, Y_m) + b\kappa(Z, Y_2, \dots, Y_m)$. For $m \geq 2$ it is unchanged when a constant is added to any argument. In particular $\kappa(cY, \dots, cY) = c^m \kappa(Y, \dots, Y)$.
- (c) **Independence.** If $\{Y_1, \dots, Y_m\}$ splits into two non-empty groups independent of one another, then $\kappa(Y_1, \dots, Y_m) = 0$. Deduce that $\kappa(Y + Z, \dots, Y + Z) = \kappa(Y, \dots, Y) + \kappa(Z, \dots, Z)$ for independent Y, Z , with m arguments throughout.
- (d) **The Gaussian is where they stop.** For $X \sim \mathcal{N}(0, \Sigma)$, $\kappa(X_a, X_b) = \Sigma_{ab}$ and every joint cumulant of order $\neq 2$ vanishes.
- (e) For centered Y , $\kappa_3(Y) = \mathbb{E}[Y^3]$ and $\kappa_4(Y) = \mathbb{E}[Y^4] - 3(\mathbb{E}[Y^2])^2$.

For (b), give Y and Z separate variables s, s' in the log-transform and apply the chain rule once. For (d), use Definition 1.3 of Lectures 1–2 with $\xi = it$.

Exercise 1.17 (TYPE C). Let $X_1, X_2, \dots$ be i.i.d. with mean μ , variance σ^2 and all moments finite, and put

$$S_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \mu).$$

Show that $\kappa(S_n) = 0$ and that, with m arguments in each cumulant,

$$\kappa(S_n, \dots, S_n) = n^{1-m/2} \kappa(X_1, \dots, X_1) \quad (m \geq 2).$$

Deduce that $\kappa(S_n, S_n) = \sigma^2$ for every n while every cumulant of order ≥ 3 tends to 0, and conclude from the moment–cumulant formula that $\mathbb{E}[S_n^m] \rightarrow \mathbb{E}[(\sigma g)^m]$ for every m , where $g \sim \mathcal{N}(0, 1)$.

This is the central limit theorem seen through cumulants: the $\sqrt{n}$ normalization is exactly the rescaling that freezes κ_2 and kills everything above it. The factor $n^{-1/2}$ in κ_3 is the rate in Theorem 2.6 of Lectures 3–4.

ASSIGNED Lecture 5, due at the start of Lecture 6: Exercises 1.9, 1.12, 1.13, 1.16, 1.17 — two A, two B, one C. If you are short of time, do the C problem.

2 The semicircle law

Q. Where are the eigenvalues of a large random symmetric matrix?

Their moments are normalized traces. A trace is a sum over closed paths, and at leading order only the paths that run along a tree contribute. Pairing the two steps on each edge of such a path gives a non-crossing pair partition, and there are Catalan-many of those.

2.1 The theorem

Recall Definition 1.7: $M = \frac{1}{\sqrt{2N}}(A + A^\top)$ with A having i.i.d. $\mathcal{N}(0,1)$ entries, so that M is symmetric, its entries $\{M_{ij}\}_{i \leq j}$ are independent centered Gaussians, and

$$\mathbb{E}[M_{ij}M_{kl}] = \frac{1}{N}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}), \quad \text{in particular} \quad \mathbb{E}M_{ij}^2 = \frac{1}{N} \ (i \neq j), \quad \mathbb{E}M_{ii}^2 = \frac{2}{N}.$$

M has N real eigenvalues $\lambda_1, \dots, \lambda_N$. Figure 1 shows their histogram.

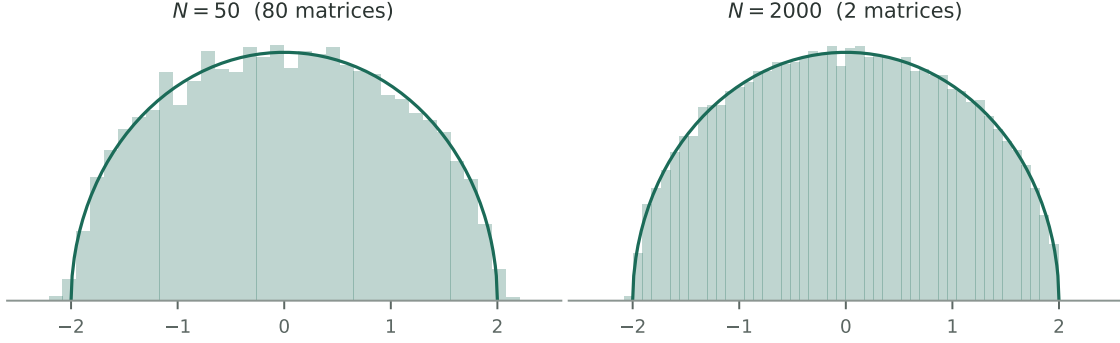


Figure 1: Histograms of the eigenvalues of Gaussian Wigner matrices, normalized to have area one, with the curve $\frac{1}{2\pi}\sqrt{4-x^2}$ drawn over them.

Theorem 2.1 (Wigner). *Let $\rho_{sc}(x) = \frac{1}{2\pi}\sqrt{4-x^2}$ on $[-2, 2]$. For every $k \geq 0$,*

$$\frac{1}{N} \sum_{i=1}^N \lambda_i^k \longrightarrow \int_{-2}^2 x^k \rho_{sc}(x) dx \quad \text{in probability.}$$

Stated without proof: the method of moments. A probability density with bounded support is determined by its moments. Consequently Theorem 2.1 implies $\frac{1}{N} \sum_i f(\lambda_i) \rightarrow \int f \rho_{sc}$ in probability for every bounded continuous f : the histogram converges to ρ_{sc} .

2.2 The strategy

The proof rests on four ideas.

1. **Moments converge to moments.** If the eigenvalues have a limiting distribution with density ρ , in the sense that $\frac{1}{N} \sum_i f(\lambda_i) \rightarrow \int f \rho$, then taking $f(x) = x^k$ the eigenvalue moments $\frac{1}{N} \sum_i \lambda_i^k$ should converge to the moments of ρ . And the eigenvalue moments are traces (Lemma 2.2).
2. **The average is a Gaussian computation plus combinatorics.** $\text{tr } M^k$ is a polynomial in the entries of M , so Theorem 1.2 computes $\mathbb{E} \text{tr } M^k$. What is left is to organize the sum, which is a counting problem (§2.3–§2.6).
3. **The fluctuations are small.** If $\text{Var}(\frac{1}{N} \text{tr } M^k) / (\frac{1}{N} \mathbb{E} \text{tr } M^k)^2 \rightarrow 0$, Chebyshev's inequality turns convergence of the mean into convergence in probability (§2.7).
4. **Identify the limit.** The limits of the means turn out to be the Catalan numbers, and these are the moments of ρ_{sc} (Proposition 2.21, Exercise 2.26).

Lemma 2.2. For every $k \geq 1$,

$$\frac{1}{N} \sum_{i=1}^N \lambda_i^k = \frac{1}{N} \operatorname{tr} M^k.$$

Proof. Diagonalize: $M = O\Lambda O^\top$ with O orthogonal and Λ diagonal, so $M^k = O\Lambda^k O^\top$ and $\operatorname{tr} M^k = \operatorname{tr} \Lambda^k = \sum_i \lambda_i^k$ by cyclicity of the trace. ■

Remark 2.3 (The empirical spectral measure). The “limiting distribution” is the limit of the probability measure $\hat{\rho}_M = \frac{1}{N} \sum_i \delta_{\lambda_i}$ that puts mass $\frac{1}{N}$ at each eigenvalue. The argument below never needs it, and works with the numbers $\frac{1}{N} \operatorname{tr} M^k$ directly. It returns in §3, Lecture 7.

2.3 Two moments by hand

Averaging $\|Mu\|^2 = u^\top M^2 u$ of Proposition 1.10 over the vectors of an orthonormal basis gives $\frac{1}{N} \operatorname{tr} M^2$, the first moment to compute.

Proposition 2.4. $\mathbb{E} \operatorname{tr} M^2 = N + 1$.

Proof. $\operatorname{tr} M^2 = \sum_{i,j} M_{ij} M_{ji}$, and (4) gives $\mathbb{E}[M_{ij} M_{ji}] = \frac{1}{N} (\delta_{ij} \delta_{ji} + \delta_{ii} \delta_{jj}) = \frac{1}{N} (1 + \delta_{ij})$, so

$$\mathbb{E} \operatorname{tr} M^2 = \frac{1}{N} \sum_{i,j} (1 + \delta_{ij}) = \frac{1}{N} (N^2 + N) = N + 1. \quad \blacksquare$$

For the fourth moment Theorem 1.2 offers three pair partitions of the four factors.

Proposition 2.5. $\mathbb{E} \operatorname{tr} M^4 = 2N + 5 + \frac{5}{N}$.

Proof. Expanding, $\operatorname{tr} M^4 = \sum_{i,j,k,l} M_{ij} M_{jk} M_{kl} M_{li}$, and Theorem 1.2 splits each term into the three pair partitions of the four factors. Take them in turn.

The pairing $\{12\}\{34\}$. By (4), $\mathbb{E}[M_{ij} M_{jk}] = \frac{1}{N} (\delta_{ij} \delta_{jk} + \delta_{ik})$ and $\mathbb{E}[M_{kl} M_{li}] = \frac{1}{N} (\delta_{kl} \delta_{li} + \delta_{ki})$. Multiplying out gives four terms; summing each over i, j, k, l counts the index tuples it allows:

$$\underbrace{\delta_{ij} \delta_{jk} \delta_{kl} \delta_{li}}_{\text{all equal: } N} + \underbrace{\delta_{ij} \delta_{jk} \delta_{ki}}_{i=j=k: N^2} + \underbrace{\delta_{ik} \delta_{kl} \delta_{li}}_{i=k=l: N^2} + \underbrace{\delta_{ik}}_{i=k: N^3},$$

so the contribution is $\frac{1}{N^2} (N^3 + 2N^2 + N) = N + 2 + \frac{1}{N}$.

The pairing $\{14\}\{23\}$. The same computation with $\mathbb{E}[M_{ij} M_{li}] = \frac{1}{N} (\delta_{il} \delta_{ij} + \delta_{jl})$ and $\mathbb{E}[M_{jk} M_{kl}] = \frac{1}{N} (\delta_{jk} \delta_{kl} + \delta_{jl})$ gives $N + 2 + \frac{1}{N}$ again.

The pairing $\{13\}\{24\}$. Here

$$\mathbb{E}[M_{ij} M_{kl}] \mathbb{E}[M_{jk} M_{li}] = \frac{1}{N^2} (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) (\delta_{jl} \delta_{ki} + \delta_{ji} \delta_{kl}),$$

whose four terms allow N^2 , N , N and N tuples respectively — the first is $\delta_{ik} \delta_{jl}$, two constraints, and each of the other three forces all four indices equal. The contribution is $\frac{1}{N^2} (N^2 + 3N) = 1 + \frac{3}{N}$.

Adding, $\mathbb{E} \operatorname{tr} M^4 = 2(N + 2 + \frac{1}{N}) + 1 + \frac{3}{N} = 2N + 5 + \frac{5}{N}$. ■

Dividing by N , $\frac{1}{N} \mathbb{E} \operatorname{tr} M^2 \rightarrow 1$ and $\frac{1}{N} \mathbb{E} \operatorname{tr} M^4 \rightarrow 2$. Two of the three pairings contributed at order N and the third only at order 1. The rest of this section explains which pairings win, and why.

Exercise 2.6 (TYPE A). Show that $\mathbb{E} \operatorname{tr} M^{2k+1} = 0$ for every $k \geq 0$, with no computation — say why. Then compute $\mathbb{E}[(\operatorname{tr} M)^2]$.

Exercise 2.7 (TYPE B). Show that $\operatorname{Var}(\operatorname{tr} M^2) = 4 + \frac{4}{N}$. Expand the square, apply Theorem 1.2 to the four-fold products, and note that the partition $\{12\}\{34\}$ reproduces $(\mathbb{E} \operatorname{tr} M^2)^2$ and is subtracted off, while the other two are equal. The remaining sum is $\frac{2}{N^2} \sum_{ijkl} (\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})^2$, and $\delta^2 = \delta$. The mean is of order N and the variance is $O(1)$: the fluctuations of $\operatorname{tr} M^2$ do not grow with N at all. That is what §2.7 needs in order to turn a statement about averages into a statement about one matrix.

Exercise 2.8 (TYPE C). Drop the Gaussian hypothesis. Let ξ_{ij} , $i \leq j$, be i.i.d. centered with variance 1 and finite fourth moment, and build a symmetric M by

$$M_{ij} = M_{ji} = \frac{\xi_{ij}}{\sqrt{N}} \quad (i < j), \quad M_{ii} = \frac{\sqrt{2}\xi_{ii}}{\sqrt{N}},$$

so that (4) still holds but Theorem 1.2 is false. Show that

$$\mathbb{E} \operatorname{tr} M^4 = 2N + 5 + \frac{5}{N} + \kappa_4(\xi) \left(1 + \frac{3}{N}\right),$$

which is Proposition 2.5 again when ξ is Gaussian, and whose leading term is $2N$ whatever ξ is. The leading order does not know what the entries are. That is universality, and with §1.5* in hand it costs one computation. Check the formula numerically with $\xi_{ij} = \pm 1$, for which $\kappa_4 = -2$.

Exercise 2.9 (TYPE B). Take $N = 2$. Show that the eigenvalues of M are $X \pm R$, where $X \sim \mathcal{N}(0, \frac{1}{2})$ and $R \geq 0$ are independent and R has density $2re^{-r^2}$. Deduce that an eigenvalue chosen uniformly at random has the law of $X + \varepsilon R$ with $\varepsilon = \pm 1$ a fair sign independent of both, and that the gap $\lambda_2 - \lambda_1$ has density $\frac{s}{2}e^{-s^2/4}$. Write $M = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ and use that $\frac{a+c}{2}$, $\frac{a-c}{2}$ and b are independent. The gap density vanishes at $s = 0$: two eigenvalues of a random symmetric matrix avoid each other. Lecture 8 explains why.

Exercise 2.10 (TYPE A). Reproduce Figure 1: plot the histogram of the eigenvalues at $N = 50$ and $N = 2000$ against ρ_{sc} , then once more with entries $\pm 1/\sqrt{N}$, which are not Gaussian and for which Theorem 1.2 is false. One figure. Universality, weeks before the course explains it.

2.4 Matrix products are sums over paths

For matrices A and B , $(AB)_{ij} = \sum_l A_{il}B_{lj}$: a sum over all ways to go from i to j in two steps, via some l , weighted by the product of the entries used. Iterating gives the same description of any power.

Definition 2.11 (Paths). A path of length m from i to j is a sequence of vertices in $\{1, \dots, N\}$,

$$\pi = (\pi(0), \pi(1), \dots, \pi(m)), \quad \pi(0) = i, \quad \pi(m) = j,$$

also written $\pi(0) \rightarrow \pi(1) \rightarrow \dots \rightarrow \pi(m)$. It is *closed* if $\pi(m) = \pi(0)$. Its a th step goes from $\pi(a-1)$ to $\pi(a)$ and *traverses* the edge $\{\pi(a-1), \pi(a)\}$, which is a loop if $\pi(a-1) = \pi(a)$. Its *weight* is

$$W(\pi) = \prod_{a=1}^m M_{\pi(a-1)\pi(a)}.$$

The entry on an edge does not depend on the direction of traversal, because $M_{ab} = M_{ba}$. Expanding the products,

$$(M^m)_{ij} = \sum_{\pi: i \rightarrow j} W(\pi), \quad \text{so} \quad \mathbb{E} \operatorname{tr} M^m = \sum_{i=1}^N \sum_{\pi: i \rightarrow i} \mathbb{E} W(\pi),$$

the inner sums running over paths of length m from i to j , respectively from i back to i . We record this as

$$\mathbb{E} \operatorname{tr} M^m = \sum_{\pi \text{ closed, length } m} \mathbb{E} W(\pi). \quad (5)$$

Lemma 2.12. *Let π be a closed path of length m . Then $\mathbb{E} W(\pi) = 0$ unless π traverses every edge, loops included, an even number of times. In every case $|\mathbb{E} W(\pi)| \leq c_m N^{-m/2}$, with c_m depending only on m .*

Proof. Group the factors of $W(\pi)$ by edge. Entries on distinct edges are independent, so $\mathbb{E} W(\pi) = \prod_e \mathbb{E}[M_e^{n_e}]$, where n_e is the number of traversals of e and $\sum_e n_e = m$. Odd moments of a centered Gaussian vanish, and $|\mathbb{E} M_e^n| \leq (2/N)^{n/2} \mathbb{E}|g|^n$. ■

In particular only closed paths that traverse every edge at least twice contribute to (5).

2.5 The same two moments, by paths

$m = 2$. A closed path of length 2 is $i \rightarrow j \rightarrow i$: it traverses the edge $\{i, j\}$ twice, out and back. There are $N(N-1)$ such paths with $i \neq j$, each with $\mathbb{E} M_{ij}^2 = \frac{1}{N}$, and N loops, each with $\mathbb{E} M_{ii}^2 = \frac{2}{N}$: again $\mathbb{E} \operatorname{tr} M^2 = N - 1 + 2 = N + 1$.

$m = 4$. Sort the closed paths of length 4 that traverse every edge at least twice by their shape:

shape	example	vertices	contribution
two edges from one vertex	$i \rightarrow j \rightarrow i \rightarrow l \rightarrow i$	3	$N(N-1)(N-2) \cdot N^{-2} = N + O(1)$
a path out and back	$i \rightarrow j \rightarrow k \rightarrow j \rightarrow i$	3	$N(N-1)(N-2) \cdot N^{-2} = N + O(1)$
one edge, four times	$i \rightarrow j \rightarrow i \rightarrow j \rightarrow i$	2	$N(N-1) \cdot 3N^{-2} = O(1)$
loops, other coincidences		≤ 2	$O(1)$

The first two shapes use three distinct vertices and two distinct edges, each traversed exactly twice, once out and once back; they are the ones of order N . The graph they trace out is a *tree*. The third uses one edge four times and has only two vertices.

From a path to a pair partition. In a path that traverses each edge exactly twice, pair the two steps that traverse the same edge. For $i \rightarrow j \rightarrow i \rightarrow l \rightarrow i$ this pairs steps 1, 2 and steps 3, 4: the partition $\{12\}\{34\}$. For $i \rightarrow j \rightarrow k \rightarrow j \rightarrow i$ it pairs steps 2, 3 and steps 1, 4: the partition $\{14\}\{23\}$. These are exactly the two pairings that contributed at order N in Proposition 2.5. The third pairing, $\{13\}\{24\}$, would need step 1 and step 3 on one edge and steps 2, 4 on another; in a closed path $i \rightarrow j \rightarrow k \rightarrow l \rightarrow i$ that forces $i = k$ and $j = l$, the single-edge path, with only two vertices. What distinguishes it is the following.

Definition 2.13 (Crossing). Draw the $2q$ points of a pair partition on a line and join each block by an arc above it. Two blocks $\{a, b\}$ and $\{c, d\}$ *cross* if $a < c < b < d$. A pair partition with no crossing blocks is *non-crossing*; there are 1, 2, 5, 14, ... of them on 2, 4, 6, 8 points, against 1, 3, 15, 105 pair partitions in total.

Remark 2.14 (What a crossing costs). A path of order N had three vertices, a path compatible with the crossing pairing only two. Each vertex is a free index and is worth a factor of N . The crossing pairing loses one vertex, and with it one power of N .

Two warnings about the correspondence. First, a path determines its pairing only at leading order: the single-edge path $i \rightarrow j \rightarrow i \rightarrow j \rightarrow i$ traverses its edge four times, so Theorem 1.2 pairs its steps in all three ways, $\mathbb{E}M_{ij}^4 = 3(\mathbb{E}M_{ij}^2)^2$, and no single pairing belongs to it. Second, “crossing” refers only to the arcs of a pair partition; the steps of a path *traverse* edges.

Remark 2.15 (Real versus complex). The two delta terms in (4) come from $M_{ij} = M_{ji}$. A complex Hermitian matrix has $\mathbb{E}[H_{ij}H_{kl}] = \frac{1}{N}\delta_{il}\delta_{jk}$, one term instead of two, and then the crossing partition leaves three constraints rather than two and costs N^{-2} . One extra way of gluing two entries together is the whole difference between the two ensembles.

2.6 Every moment

The odd moments vanish: $\mathbb{E} \operatorname{tr} M^{2k+1} = 0$, because the weight of a path of odd length is a product of an odd number of centered jointly Gaussian variables. From now on the power is even, $2k$.

Fix a starting vertex. Every closed path starts somewhere, and the starting point can be fixed at no cost.

Lemma 2.16. *For every vertex i ,*

$$\frac{1}{N} \mathbb{E} \operatorname{tr} M^{2k} = \mathbb{E}(M^{2k})_{ii} = \sum_{\pi: i \rightarrow i} \mathbb{E} W(\pi).$$

Proof. For a permutation matrix Q , the matrix $QMQ^\top$ has the same law as M (Remark 1.8), and $(QM^{2k}Q^\top)_{ii}$ is another diagonal entry of M^{2k} . So all N diagonal entries of M^{2k} have the same law, and their average $\frac{1}{N} \operatorname{tr} M^{2k}$ has the same mean as any one of them. ■

Fixing the starting vertex removes the factor $\frac{1}{N}$, and the whole argument is now a count of vertices. By Lemma 2.12 with $m = 2k$, $\mathbb{E} W(\pi) = 0$ unless π traverses every edge an even number of times, and $|\mathbb{E} W(\pi)| \leq c_{2k} N^{-k}$ always.

Lemma 2.17 (A tree is the sparsest connected graph). *Let π be a path of length $2k$ from i to i that traverses every edge at least twice, and let v be its number of distinct vertices, i included. Then $v \leq k + 1$, with equality if and only if the edges it traverses form a tree with k edges, each traversed exactly twice.*

Proof. Each edge takes at least two of the $2k$ steps, so there are at most k distinct edges. The graph they form is connected, because a single path runs through all of it, and a connected graph on v vertices has at least $v - 1$ edges that are not loops, with equality exactly for a tree. So $v \leq k + 1$, and equality forces k distinct edges, none a loop, forming a tree, each traversed exactly twice. ■

Call a path as in the equality case a *tree path*. Deleting an edge disconnects a tree, so a path in a tree that returns to its start traverses each edge alternately in each direction: a tree path traverses each of its k edges once out and once back.

Proposition 2.18 (Only tree paths survive). *Let $T_{k,N}$ be the number of tree paths of length $2k$ from i . Then*

$$\frac{1}{N} \mathbb{E} \operatorname{tr} M^{2k} = N^{-k} T_{k,N} + O\left(\frac{1}{N}\right).$$

Proof. Sort the paths $\pi : i \rightarrow i$ by their number v of distinct vertices. By Lemma 2.12, only paths traversing every edge at least twice contribute, and by Lemma 2.17 these have $v \leq k + 1$.

Paths with v vertices are at most $c_{k,v} N^{v-1}$ in number. Record which of the times $0, 1, \dots, 2k$ visit the same vertex: there are at most $c_{k,v}$ such patterns, depending only on k and v . Given the pattern, the $v - 1$ vertices other than i are chosen from $N - 1$.

So the paths with $v \leq k$ contribute at most $c_{k,v} N^{v-1} \cdot c_{2k} N^{-k} = O(1/N)$ in total, by Lemma 2.12. The paths with $v = k + 1$ are the tree paths. A tree has no loops, so each of the k edges of a tree path joins two distinct vertices, carries an entry of variance $\frac{1}{N}$, and is traversed exactly twice: $\mathbb{E}W(\pi) = N^{-k}$ exactly. $\blacksquare$

It remains to count tree paths, and this is where the non-crossing pair partitions enter. The *pairing* of a tree path pairs the two steps that traverse each edge.

Lemma 2.19. *The pairing of a tree path is non-crossing. Conversely, each non-crossing pair partition of the $2k$ steps is the pairing of exactly $(N - 1)(N - 2) \cdots (N - k)$ tree paths from i .*

Proof. Suppose the pairing of a tree path had crossing blocks $\{a, b\}$ and $\{c, d\}$, $a < c < b < d$, and let e be the edge traversed at steps a and b . Between steps a and b the path is on the far side of e , and after step b it is on the near side for good, since e is traversed only twice. The edge traversed at step c therefore lies on the far side, yet it is traversed again at step d , from the near side. That is impossible.

For the converse, induct on k . For $k = 1$ the tree paths are $i \rightarrow j \rightarrow i$ with $j \neq i$, and there are $N - 1$ of them. A non-crossing pair partition P has a block of adjacent steps $\{r, r + 1\}$: take a block $\{a, b\}$ with $b - a$ smallest, and note that a point strictly inside it would be paired inside it. In a tree path with pairing P , steps r and $r + 1$ traverse the same edge out and back, so $\pi(r + 1) = \pi(r - 1)$, and $\pi(r)$ is visited at no other time: it is a leaf, and it is not i . Deleting those two steps leaves a tree path of length $2k - 2$ from i whose pairing is P with that block removed, still non-crossing. Conversely, any such shorter tree path extends by inserting an excursion to a new leaf after its $(r - 1)$ st step, and the leaf can be any of the $N - k$ vertices it has not yet used. So the count is $(N - 1) \cdots (N - k + 1) \cdot (N - k)$. $\blacksquare$

Proposition 2.20. *Let C_k be the number of non-crossing pair partitions of $2k$ points. Then*

$$\frac{1}{N} \mathbb{E} \operatorname{tr} M^{2k} = C_k \left(1 - \frac{1}{N}\right) \left(1 - \frac{2}{N}\right) \cdots \left(1 - \frac{k}{N}\right) + O\left(\frac{1}{N}\right) \rightarrow C_k.$$

Proof. By Lemma 2.19, $T_{k,N} = C_k(N - 1) \cdots (N - k)$. Multiply by N^{-k} and apply Proposition 2.18. $\blacksquare$

Proposition 2.21 (Catalan). *Let C_k be the number of non-crossing pair partitions of $2k$ points, with $C_0 = 1$. Then $C_k = \sum_{j=0}^{k-1} C_j C_{k-1-j}$, so $C_1, \dots, C_5 = 1, 2, 5, 14, 42$. The generating function $C(w) = \sum_k C_k w^k$ satisfies*

$$C(w) = 1 + w C(w)^2, \tag{6}$$

so that $C(w) = \frac{1 - \sqrt{1 - 4w}}{2w}$ and $C_k = \frac{1}{k+1} \binom{2k}{k}$.

Proof. Let $\{1, b\}$ be the block containing the point 1. A block joining a point strictly between 1 and b to a point after b would cross $\{1, b\}$, so the points strictly inside pair among themselves and those after b pair among themselves. Hence $b = 2j + 2$ for some j , the inside carries any of C_j partitions and the outside any of C_{k-1-j} , and summing over j gives the recursion. Multiplying by w^k and summing over $k \geq 1$ turns the convolution into a product, $C - 1 = wC^2$. Of the two roots, only $\frac{1-\sqrt{1-4w}}{2w}$ is finite at $w = 0$, and the binomial series for $\sqrt{1-4w}$ gives the closed form. ■

2.7 The fluctuations are small

Proposition 2.22. *For every m , $\text{Var}(\frac{1}{N} \text{tr } M^m) = O(1/N)$.*

Proof. By (5), $\text{Var}(\text{tr } M^m) = \sum_{\pi, \pi'} \text{Cov}(W(\pi), W(\pi'))$ over pairs of closed paths of length m . If π and π' traverse no common edge, $W(\pi)$ and $W(\pi')$ are functions of disjoint sets of entries, hence independent, and the covariance is 0. If some edge is traversed an odd number of times by the two paths together, then $\mathbb{E}[W(\pi)W(\pi')] = 0$, and one of $\mathbb{E}W(\pi)$, $\mathbb{E}W(\pi')$ is 0, by the argument of Lemma 2.12. So the surviving pairs share an edge, which makes the union of the two paths connected, and traverse every edge at least twice in total, which gives the union at most m edges. By the argument of Lemma 2.17 it then has at most $m + 1$ vertices. There are at most cN^{m+1} such pairs, each covariance is at most $c'N^{-m}$ by Lemma 2.12, and so $\text{Var}(\text{tr } M^m) = O(N)$. ■

Since $\frac{1}{N} \mathbb{E} \text{tr } M^{2k} \rightarrow C_k > 0$, this gives $\text{Var}/(\mathbb{E})^2 \rightarrow 0$, which is idea 3. The bound is not sharp: in fact $\text{Var}(\text{tr } M^m) = O(1)$, as Exercise 2.7 shows for $m = 2$ and Exercise 2.28 for every m .

Proof of Theorem 2.1. By Lemma 2.2 and Propositions 2.20 and 2.21, the mean of $\frac{1}{N} \sum_i \lambda_i^m$ tends to C_k if $m = 2k$ and is 0 if m is odd. By Exercise 2.26 these are the moments of ρ_{sc} . By Proposition 2.22 and Chebyshev's inequality, $\frac{1}{N} \sum_i \lambda_i^m$ is within $o(1)$ of its mean with probability tending to one. ■

Remark 2.23 (What was used). Gaussianity entered only through Lemma 2.12. The proof of Proposition 2.18 used only that an edge traversed once gives 0 and that moments are bounded, and a tree path sees nothing about an entry but its variance. The same proof works for any symmetric matrix with independent centered entries, variance $\frac{1}{N}$ off the diagonal, and $\mathbb{E}|M_{ij}|^p \leq c_p N^{-p/2}$. Exercise 2.8 is the case $\text{tr } M^4$, and Exercise 2.10 the picture.

Exercise 2.24 (TYPE A). Draw all 15 pair partitions of 6 points, mark the crossings, and check that exactly 5 are non-crossing. For each of those five, draw a tree path with that pairing and check that it has 4 vertices.

Exercise 2.25 (TYPE A). Theorem 2.1 predicts $\frac{1}{N} \mathbb{E} \text{tr } M^6 \rightarrow 5$. Estimate $\frac{1}{N} \mathbb{E} \text{tr } M^6$ by simulation at $N = 50, 200, 800$, and estimate the constant c in $\frac{1}{N} \mathbb{E} \text{tr } M^6 \approx 5 + \frac{c}{N}$.

Exercise 2.26 (TYPE B). Show that ρ_{sc} is a probability density, that its odd moments vanish, and that

$$\int_{-2}^2 x^{2k} \rho_{sc}(x) dx = \frac{1}{k+1} \binom{2k}{k}.$$

Substitute $x = 2 \sin \theta$ and evaluate the resulting integral with the Beta function, or by integrating by parts to get a recursion in k .

Exercise 2.27 (TYPE B). Let H be complex Hermitian with $\{H_{ij}\}_{i < j}$ independent complex Gaussians with $\mathbb{E}|H_{ij}|^2 = \frac{1}{N}$ and $\mathbb{E}H_{ij}^2 = 0$, and $H_{ii} \sim \mathcal{N}(0, \frac{1}{N})$ real, all independent. Show that Propositions 2.20 and 2.22 hold for H , with the same leading term, so the eigenvalues of H obey Theorem 2.1 too. *In a tree path each edge is traversed once each way, so it contributes $\mathbb{E}[H_{ab}H_{ba}] = \mathbb{E}|H_{ab}|^2$. Where, then, does Remark 2.15 show up?*

Exercise 2.28 (TYPE C). Sharpen Proposition 2.22: show that $\text{Var}(\text{tr } M^m) = O(1)$, and hence $\text{Var}(\frac{1}{N} \text{tr } M^m) = O(N^{-2})$. *The proof bounded the number of vertices by $m+1$. Show that equality is impossible.*

ASSIGNED Lecture 6, due at the start of Lecture 7: Exercises 2.24, 2.10, 2.26, 2.9, 2.8, 2.28 — two A, two B, two C. Exercises 2.6, 2.7, 2.25 and 2.27 are pool problems: available and examinable, not collected. *If you are short of time, do the C problems.*

3 The semicircle law by the Stieltjes transform

Q. The counting proof produced a list of numbers, and then recognized them. Where does the density $\frac{1}{2\pi}\sqrt{4-x^2}$ come from?

From a fixed-point equation. Averaging $\frac{1}{x-z}$ over the eigenvalues gives a function of z that satisfies $m^2 + zm + 1 = 0$ in the limit, and the density can be read off from m .

A second proof of Theorem 2.1, which produces the density rather than recognizing a sequence of numbers. It needs the object that Remark 2.3 set aside.

3.1 The empirical spectral measure

Write $\langle \mu, f \rangle = \int f d\mu$ for a measure μ and a bounded continuous f .

Definition 3.1 (Point mass). For $x \in \mathbb{R}$, the *point mass* δ_x is the probability measure determined by

$$\langle \delta_x, f \rangle = f(x) \quad \text{for every bounded continuous } f.$$

It puts mass 1 at the single point x and none anywhere else. It has no density: there is no function p with $\int fp = f(x)$ for all f , since such a p would have to vanish off $\{x\}$ and still integrate to 1. Measures are the right objects here precisely because we want to place mass at points.

Definition 3.2 (Empirical spectral measure). For a symmetric $N \times N$ matrix M with eigenvalues $\lambda_1, \dots, \lambda_N$,

$$\hat{\rho}_M = \frac{1}{N} \sum_{i=1}^N \delta_{\lambda_i}, \quad \text{so} \quad \langle \hat{\rho}_M, f \rangle = \frac{1}{N} \sum_{i=1}^N f(\lambda_i).$$

It is a probability measure: total mass $\frac{1}{N} \cdot N = 1$. It records the eigenvalues and nothing else — not which eigenvector goes with which, not the order.

Now let M be random. Then $\hat{\rho}_M$ is a *random measure*: for each fixed f , the number $\langle \hat{\rho}_M, f \rangle$ is a random variable. There is one sensible way to average it.

Definition 3.3 (Mean measure). $\mathbb{E}\hat{\rho}$ is the measure determined by

$$\langle \mathbb{E}\hat{\rho}, f \rangle = \mathbb{E} \langle \hat{\rho}_M, f \rangle \quad \text{for every bounded continuous } f.$$

The left side is a number attached to a deterministic measure; the right side is the expectation of a random variable. Nothing is being integrated twice — the definition simply says that averaging a measure means averaging what it does to each test function.

Lemma 3.4. $\mathbb{E}\hat{\rho}$ is the law of λ_J , where J is uniform on $\{1, \dots, N\}$ and independent of M : it is the distribution of one eigenvalue picked at random.

Proof. $\mathbb{E}[f(\lambda_J)] = \mathbb{E}[\frac{1}{N} \sum_i f(\lambda_i)] = \mathbb{E}\langle \hat{\rho}_M, f \rangle$. ■

Remark 3.5 (Two levels of randomness, kept apart). $\hat{\rho}_M$ is random; $\mathbb{E}\hat{\rho}$ is not. A statement about $\mathbb{E}\hat{\rho}$ is a statement about averages over many independent matrices. A statement about $\hat{\rho}_M$ is a statement about one matrix, and getting from the first to the second is exactly a variance estimate. Both appear below, and they are not the same theorem.

3.2 What the counting proof cannot do

- (i) It never produced a **density**. It produced a sequence of numbers that we recognized; perturb the model and there is nothing to recognize.
- (ii) It requires **all moments finite**.
- (iii) It gives **no local information** — nothing about one eigenvalue, a gap, or the edge.
- (iv) It does not survive **deformation**: unequal variances, or a rank-one spike, and the combinatorics does not adapt.

All four are repaired by replacing the polynomials x^k with a different family of test functions.

3.3 The Stieltjes transform

Definition 3.6. For a probability law μ on $\mathbb{R}$ and $z \in \mathbb{C}$ with $\Im z > 0$,

$$m_\mu(z) = \int \frac{d\mu(\lambda)}{\lambda - z} = \left\langle \mu, \frac{1}{x-z} \right\rangle.$$

The test function $\frac{1}{x-z}$ is bounded and continuous for each fixed z off the real axis, so Definition 3.6 makes sense for every probability law, with no moment hypothesis at all. It is analytic on the upper half plane, satisfies $|m_\mu(z)| \leq 1/\Im z$, and $m_\mu(z) = -1/z + O(|z|^{-2})$ as $z \rightarrow \infty$ — the last of which will pin down a branch.

Stieltjes inversion. If μ has a continuous density ρ near λ then $\rho(\lambda) = \frac{1}{\pi} \lim_{\varepsilon \downarrow 0} \Im m_\mu(\lambda + i\varepsilon)$. We use this and do not prove it.

The moment method has no analogue of this step: it is what turns a transform back into a picture of the spectrum. The bridge to matrices is immediate from Definition 3.2, since $\frac{1}{\lambda_i - z}$ are the eigenvalues of $(M - z)^{-1}$:

$$m_{\hat{\rho}_M}(z) = \frac{1}{N} \text{tr} (M - z)^{-1}. \tag{7}$$

The matrix $(M - z)^{-1}$ is the *resolvent*. It exists for every z off the real axis because M is symmetric, and $\|(M - z)^{-1}\| \leq 1/\Im z$.

Remark 3.7. Ridge regression in §2.5 of Lectures 1–2 is $(H^\top H + \lambda I)^{-1} H^\top y$, and the matrix inverted there is (7) at $z = -\lambda$.

3.4 The self-consistent equation

Write $m_N(z) = \frac{1}{N} \mathbb{E} \operatorname{tr}(M - z)^{-1}$. Let $M^{(1)}$ be M with its first row and column deleted, and let $a \in \mathbb{R}^{N-1}$ be the first column of M with its first entry deleted. The strategy is to compute one diagonal entry of the resolvent in terms of the same object one dimension down.

Lemma 3.8 (Schur complement). $((M - z)^{-1})_{11} = \frac{1}{M_{11} - z - a^\top (M^{(1)} - z)^{-1} a}$.

This is the block-inversion identity behind the Schur complement in §2.4 of Lectures 1–2.

Now estimate the three pieces. Put $R = (M^{(1)} - z)^{-1}$, which is a function of $M^{(1)}$ alone and therefore independent of a , and note $\mathbb{E}[a_r a_s] = \delta_{rs}/N$.

- (a) $M_{11} = O_{\mathbb{P}}(N^{-1/2})$, since $\operatorname{Var} M_{11} = 2/N$.
- (b) $\mathbb{E}[a^\top R a \mid M^{(1)}] = \frac{1}{N} \operatorname{tr} R$, and its conditional variance is $\frac{2}{N^2} \operatorname{tr}(R^2)$. Both are Example 1.6 with $A = R$ and $\Sigma = \frac{1}{N} I$; the computation there is algebraic, so it applies verbatim to the complex symmetric R . Since $\|R\| \leq (\Im z)^{-1}$ we have $|\operatorname{tr}(R^2)| \leq N(\Im z)^{-2}$, so the variance is $O(1/N)$. This is the only place probability enters.
- (c) $\frac{1}{N} \operatorname{tr} R$ differs from $m_N(z)$ by $O(1/N)$, since deleting one row and column of an $N \times N$ matrix moves a normalized resolvent trace by that much.

All diagonal entries of $(M - z)^{-1}$ have the same law, so $m_N(z) = \mathbb{E}[(M - z)^{-1}]_{11}$. Feeding (a)–(c) into Lemma 3.8 and passing to the limit,

$$\boxed{m(z) = \frac{1}{-z - m(z)}, \quad \text{that is} \quad m^2 + zm + 1 = 0.} \quad (8)$$

Asserted. That $\frac{1}{N} \operatorname{tr}(M - z)^{-1}$ concentrates around its mean, so that the limit in (8) may be taken inside, is a concentration estimate we do not prove.

Equation (8) is a fixed-point equation for the answer, derived without knowing the answer. The moment method had to be told what to recognize.

3.5 Solving, and the comparison

Both roots of (8) are $\frac{-z \pm \sqrt{z^2 - 4}}{2}$. As $z \rightarrow \infty$, $\sqrt{z^2 - 4} = z - 2/z + O(|z|^{-3})$, so only the $+$ root satisfies $m \sim -1/z$:

$$m(z) = \frac{-z + \sqrt{z^2 - 4}}{2}.$$

For $\lambda \in (-2, 2)$ we have $\sqrt{\lambda^2 - 4} = i\sqrt{4 - \lambda^2}$, so $\Im m(\lambda + i0) = \frac{1}{2}\sqrt{4 - \lambda^2}$, and Stieltjes inversion returns $\rho_{sc}(\lambda) = \frac{1}{2\pi}\sqrt{4 - \lambda^2}$. That is Theorem 2.1 again, now with its density.

Remark 3.9 (The two proofs are the same equation). Expanding Definition 3.6 for large $|z|$ gives $m(z) = -\sum_{k \geq 0} C_k z^{-(2k+1)}$, so with $w = z^{-2}$ we have $-zm(z) = C(w)$. Substituting into the Catalan relation (6) returns $-zm = 1 + m^2$, which is (8). *The recursion that counts non-crossing pair partitions and the self-consistent equation for the resolvent are one equation written twice.*

Return to the four complaints above. The density came out directly. Only the second moment of the entries was used, so nothing needed all moments. Unequal variances $\mathbb{E}M_{ij}^2 = S_{ij}/N$ turn (8) into a system of N coupled equations rather than one, which is a change of size and not of method. A rank-one spike is a rank-one perturbation of the resolvent. And shrinking $\Im z$ toward the real axis, rather than holding it fixed, is precisely what converts a global law into a local one.

Exercise 3.10 (TYPE A). Compute m_μ for $\mu = \delta_0$, for $\mu = \frac{1}{2}(\delta_{-1} + \delta_1)$, and for μ uniform on $[0, 1]$. In each case check $|m_\mu(z)| \leq 1/\Im z$ and $m_\mu(z) = -1/z + O(|z|^{-2})$, and recover the density of the third by Stieltjes inversion.

Exercise 3.11 (TYPE A). Numerically, for a GOE matrix at $N = 2000$, plot $\frac{1}{\pi} \Im \left[\frac{1}{N} \operatorname{tr}(M - \lambda - i\varepsilon)^{-1} \right]$ against $\lambda \in [-3, 3]$ for $\varepsilon = 1, 0.1, 0.01, 0.001$, with ρ_{sc} overlaid. *At $\varepsilon = 1$ the curve is a smoothed semicircle; at $\varepsilon = 0.001$ it is a forest of spikes at the individual eigenvalues. Somewhere in between is the global law. That crossover is at $\varepsilon \sim 1/N$, and it is the whole content of the word “local”.*

Exercise 3.12 (TYPE B). Prove Remark 3.9 carefully. Expand $\frac{1}{\lambda - z} = -\sum_{k \geq 0} \lambda^k z^{-(k+1)}$ for $|z| > 2$, integrate against ρ_{sc} to get $m(z) = -\sum_k C_k z^{-(2k+1)}$ (odd moments vanish), substitute $w = z^{-2}$ to identify $-zm(z) = C(w)$, and check that $C = 1 + wC^2$ is (8).

Exercise 3.13 (TYPE B). Prove estimate (c): if M is symmetric $N \times N$ and $M^{(1)}$ deletes one row and column, then

$$\left| \operatorname{tr}(M - z)^{-1} - \operatorname{tr}(M^{(1)} - z)^{-1} \right| \leq \frac{\pi}{\Im z}.$$

Cauchy interlacing gives $\lambda_i \leq \mu_i \leq \lambda_{i+1}$ for the eigenvalues of M and $M^{(1)}$, so the two eigenvalue counting functions differ by at most 1 at every point. Write both traces as integrals against those counting functions, integrate by parts, and use $\int_{\mathbb{R}} |x - z|^{-2} dx = \pi/\Im z$. Note that the bound does not depend on N , which is why dividing by N makes it $O(1/N)$.

Exercise 3.14 (TYPE C). *Unequal variances.* Let M be symmetric Gaussian with $\mathbb{E}M_{ij}^2 = S_{ij}/N$, where S is a fixed symmetric matrix with non-negative entries, and let $m_i(z) = \mathbb{E}[(M - z)^{-1}]_{ii}$. Redo the derivation above with the i th row and column deleted instead of the first, and derive the system

$$m_i(z) = \frac{1}{-z - \frac{1}{N} \sum_j S_{ij} m_j(z)}, \quad i = 1, \dots, N.$$

Check that $S_{ij} \equiv 1$ returns (8). *Then take S block-constant with two blocks and solve the resulting pair of equations; the spectrum is no longer a semicircle, and no combinatorial recognition step could have produced it.*

ASSIGNED Lecture 7, due at the start of Lecture 8: Exercises 3.10–3.14 — two A, two B, one C. *If you are short of time, do the C problem.*

Lecture 8 derives the exact joint law of the eigenvalues and reads it as a gas of repelling charges; the semicircle appears a third time, as its equilibrium measure.